Zero Shot Sentiment Analysis on Tweets in Any Language

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Objective

We train a sentiment analysis convolutional neural network classification model on a set of sentiment-labeled tweets in a popular language, and aim to transfer learn to classify sentiment in any other language given a set of unlabeled tweets in that language.

Introduction

We re-purpose the Unicode Convolutional Neural Network [1] which offers two advantages.

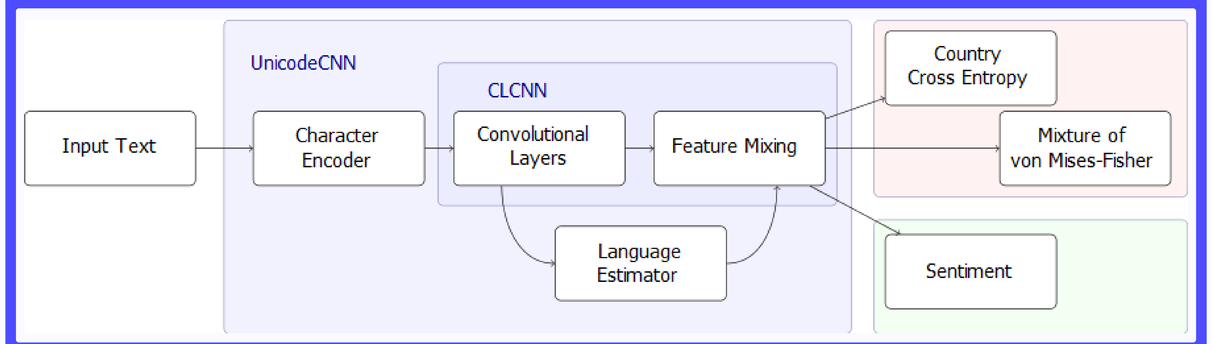
1. It is pre-trained to solve the geolocation problem, which has massive amounts of reliable data.
2. It is already capable of analyzing text in any language by looking at its underlying representation in Unicode.

Below are example tweets which exploit similarity in location and Unicode to train the model to transfer learn sentiment from English to Portuguese.

Example Tweets

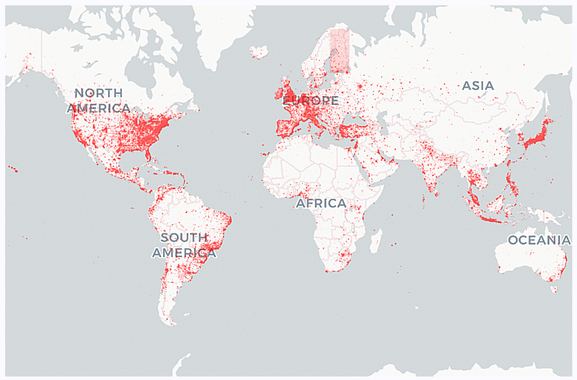


UnicodeCNN



Dataset

The majority of the natural language processing community has focused on analyzing English text since it comprises roughly 42% of all activity on Twitter, which is unfortunate since there are over 100 languages actively used on the platform. The locations of the tweets used to pre-train the UnicodeCNN on geolocation are shown below [1].

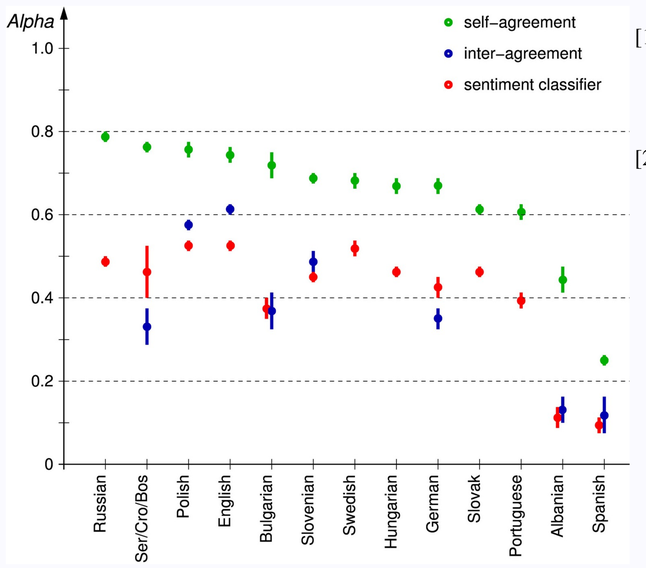


For the sentiment problem, we train using a publicly available sentiment-labeled dataset containing 1.6 million tweets in 13 languages, as shown on right [2]. While most of that data was labeled precisely, the labels for Albanian and Spanish tweets were not. Generating a precisely-labeled sentiment data for tweets of the remaining 90+ actively used languages would be impractical.

Model

In this experiment, we will use the UnicodeCNN pretrained to predict the exact GPS coordinates of any tweet in any language sent from any location by analyzing the contents at the Unicode level. It was trained on all 900 million geo-encoded tweets sent over 9 months by over 3.0 million users in over 100 languages, and replace the final MvMF estimation layer with a layer of sentiment classification as shown above.

We will retrain some layers of the model on a training set and evaluate on a testing set, where the languages of the two sets may or may not agree.



Contribution

In the main experiments highlighted in the table below, we train on sentiment-labeled English tweets, test on unlabeled Portuguese tweets, and validate against a model trained on sentiment-labeled Portuguese tweets.

#	Retrain	Train	Test
1	All	English	English
2	All	Portuguese	Portuguese
3	All	English	Portuguese
4	Final	English	English
5	Final	Portuguese	Portuguese
6	Final	English	Portuguese

We chose English and Portuguese so that we could train the main model and the validation model with high-quality data, but our tests did not require the Portuguese sentiment-labels to classify sentiment on Portuguese tweets.

After finalizing the results on Portuguese, we would like to test and validate this model on tweets in other languages. If successful, we would expect the English-trained model to succeed on other low resource languages for which significant amounts of sentiment-labeled data do not exist.

References

[1] Mike Izbicki, Evangelos Papalexakis, and Vassilis Tsotras.
Geolocating tweets in any language at any location.
CIKM, 2019.

[2] Igor Mozetič, Miha Grčar, and Jasmina Smailović.
Multilingual twitter sentiment classification: The role of human annotators.
PLOS ONE, 11(5):1–26, 05 2016.

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